



LIBRARY MANAGEMENT

SQL PROJECT

LIBRARY MANAGEMENT

Description: The Library Management System (LMS) database is designed to store, manage, and analyze data related to books, authors, members, borrowings, and other associated information. This structure enables efficient querying, tracking, and data analysis to gain insights into library operations, popular books, member activities, and resource availability.

TABLES IN THE DATABASE

1. Books Table – Stores details of each book such as title, author, genre, published year, and available copies.
2. Authors Table – Contains information about book authors including their ID, name, and country.
3. Members Table – Holds details of library members like name, contact information, and membership date.
4. Borrow Table – Tracks borrowing records including which member borrowed which book, along with borrow and return dates.

QUERIES:

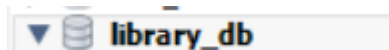
CREATING DATABASE:

Question: -Creating Database

Input Query:

```
create database Library_DB;
```

Output:



Insight: A separate database is created to manage library operations.

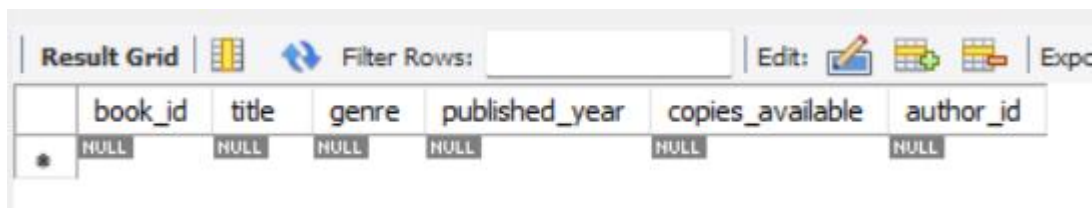
CREATE TABLE:

Question: Create a table of books where there are book id, title, genre, published year, copies available with appropriate constraints

Input Query:

```
-- Books Table
CREATE TABLE Books (
    book_id INT PRIMARY KEY AUTO_INCREMENT,
    title VARCHAR(200) NOT NULL,
    genre VARCHAR(50),
    published_year INT,
    copies_available INT,
    author_id INT,
    FOREIGN KEY (author_id) REFERENCES Authors(author_id)
);
```

Output:



	book_id	title	genre	published_year	copies_available	author_id
*	NULL	NULL	NULL	NULL	NULL	NULL

Insight: Ensures unique book ID and stores book details with author id as a foreign key.

INSERTING VALUES:

Question: Inserting values in each column.

Input Query:

```
-- insert values in books
INSERT INTO Books (title, genre, published_year, copies_available, author_id) VALUES
('Malgudi Days', 'Fiction', 1943, 5, 1),
('The Guide', 'Novel', 1958, 3, 1),
('Five Point Someone', 'Fiction', 2004, 7, 2),
('2 States', 'Romance', 2009, 8, 2),
('The Room on the Roof', 'Children', 1956, 4, 3),
('The Immortals of Meluha', 'Mythology', 2010, 6, 4),
('The God of Small Things', 'Fiction', 1997, 5, 5),
('Midnight's Children', 'Historical', 1981, 3, 6),
('Interpreter of Maladies', 'Short Stories', 1999, 6, 7),
```

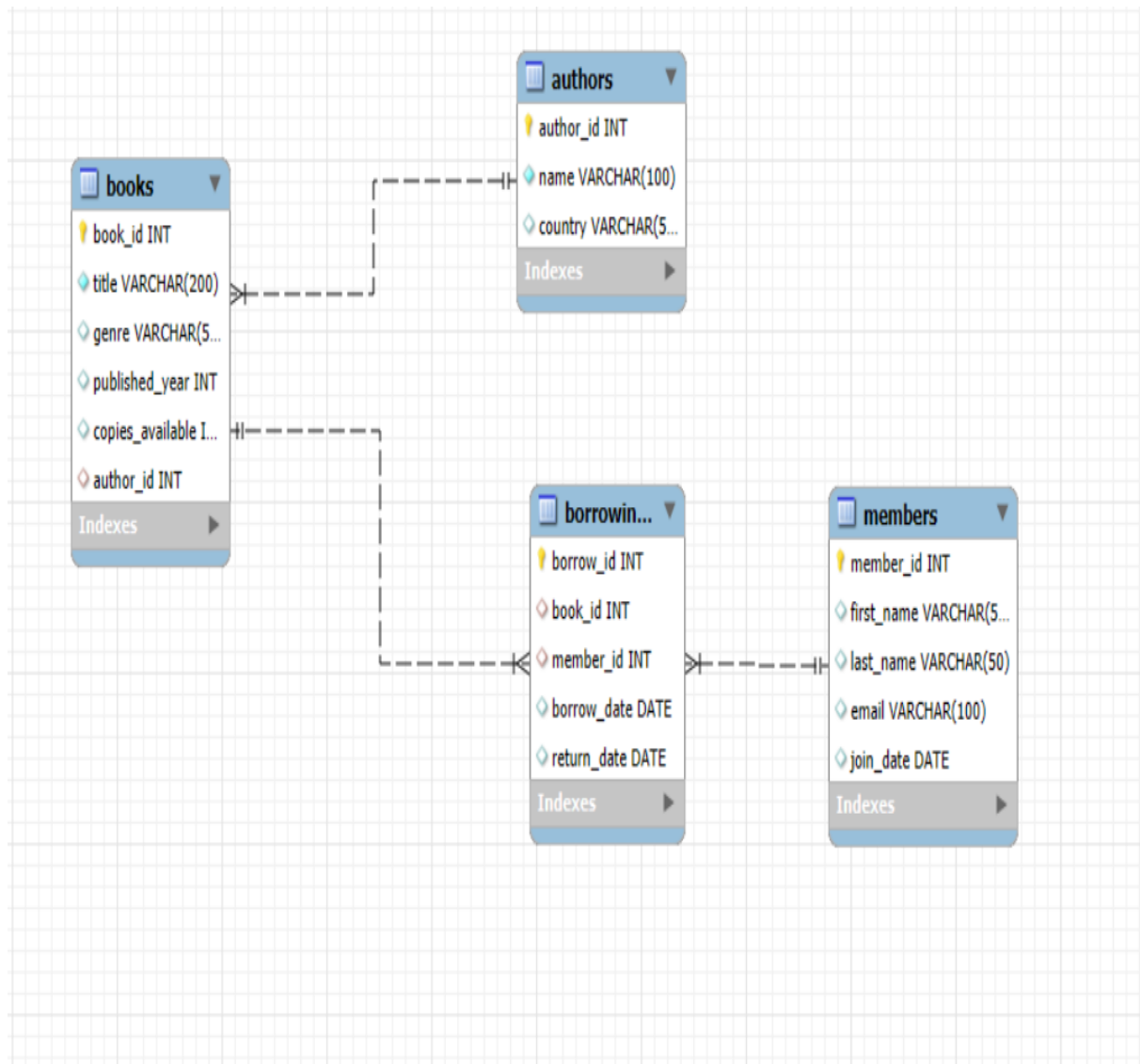
Output:

LIBRARY MANAGEMENT

	book_id	title	genre	published_year	copies_available	author_id
▶	1	Malgudi Days	Fiction	1943	5	1
	2	The Guide	Novel	1958	3	1
	3	Five Point Someone	Fiction	2004	7	2
	4	2 States	Romance	2009	8	2
	5	The Room on the Roof	Children	1956	4	3
	6	The Immortals of Meluha	Mythology	2010	6	4
	7	The God of Small Things	Fiction	1997	5	5
	8	Midnight's Children	Historical	1981	3	6
	9	Interpreter of Maladies	Short Stories	1999	6	7
	10	A Suitable Boy	Romance	1993	4	8

Insight: Data added for testing queries.

ENTITY RELATIONSHIP DIAGRAM:



DISABLE SAFE MODE

DISABLE SAFE MODE

```
SET SQL_SAFE_UPDATES=0;
```

Output:

```
1 21:30:21 set sql_safe_updates=0 0 row(s) affected
```

Update

Question: updating the borrowing table by replacing the year '2025' to null

Input Query

```
UPDATE borrowings
SET return_date = NULL
WHERE YEAR(return_date) = 2025;
```

Output:

	borrow_id	book_id	member_id	borrow_date	return_date
▶	301	107	202	2024-07-01	2024-07-15
	302	115	215	2024-02-20	2024-03-05
	303	101	115 04	2024-06-01	2024-06-15
	304	110	207	2022-04-10	2022-04-25
	305	104	211	2023-03-15	2023-03-30
	306	117	201	2025-03-20	NULL
	307	109	213	2023-07-20	2023-08-05
	308	103	210	2023-01-25	NULL
	309	112	205	2021-09-20	2021-10-05
	310	108	214	2023-10-05	2023-10-20
	311	102	206	2025-02-25	NULL
	312	111	209	2022-11-20	2022-12-05
	---	---	---	---	---

Insight: Demonstrates date manipulation.

AGGREGATE FUNCTION:

Question: finding total members.

Input Query

```
SELECT COUNT(*) AS TOTAL_MEMBERS FROM MEMBERS;
```

Output:

	TOTAL_MEMBERS
▶	15

Insight: Counts the total number of registered members in the library.

Question: Finding the Average, Min, Max copies of books available

Input Query

```
select round(avg(copies_available)) as avg_copies,
min(copies_available) as min_copies,
max(copies_available) as max_copies
from books;
```

Output

	avg_copies	min_copies	max_copies
▶	4	2	7

Insight: Shows the average, least, and highest available copies among all books that are useful for stock tracking.

DISTINCT:

Question: find all the unique genre of the books

Input Query

```
select distinct genre from books;
```

Output:

	genre
►	Fantasy
	Fiction
	Mystery
	Philosophy
	Thriller
	Classic
	Drama
	Horror
	Romance
	Historical
	Magical R...

Insight: Finds unique genre allows for analyses.

STRING FUNCTION

Input Query

```
SELECT UPPER(name) as NAME from authors;
```

Output:

	NAME
►	J.K. ROWLING
	GEORGE R.R. MARTIN
	CHETAN BHAGAT
	AGATHA CHRISTIE
	PAULO COELHO
	DAN BROWN
	HARPER LEE
	J.R.R. TOLKIEN
	ARUNDHATI ROY
	KHALED HOSSEINI
	STEPHEN KING
	MARK TWAIN
	JANE AUSTEN
	LEO TOLSTOY
	RUSKIN BOND

Insight: Converts all book titles to uppercase that demonstrates string formatting and can be useful for standardizing text.

DATE FUNCTION

Question: finding the number of years the members have joined membership.

Input Query

```
select concat(first_name," ",last_name) as full_name,
(year(current_date())- year(join_date)) as years_joined from members;
```

Output:

	full_name	years_joined
►	Amit Sharma	5
	Priya Verma	0
	Rohan Mehta	4
	Sneha Kapoor	4
	Vikram Singh	4
	Anjali Nair	3
	Rahul Desai	3
	Neha Iyer	0
	Karan Patel	3
	Divya Malhotra	2
	Siddharth Rao	2
	Pooja Sharma	2
	Ariun Khanna	2

Insight: Calculates how many years each member has been in the library since their membership start date.

Math Functions

Input Query:

```
SELECT ABS(-1000) AS absolute_result,
floor(8.3) as floor_result,
ceil(8.3) as Ceil_result,
floor(-8.3) as floor_result,
ceil(-8.3) as ceil_result,
power(4,4) as power_result,
sqrt(625) as square_result;
```

Output:

	absolute_result	floor_result	Ceil_result	floor_result	ceil_result	power_result	square_result
▶	1000	8	9	-9	-8	256	25

WHERE CLAUSE

Question: Retrieve all books where the genre is fiction and is released before 2000

Input Query

```
SELECT * FROM BOOKS
WHERE GENRE="fiction" and published_year<2000;
```

Output

	book_id	title	genre	published_year	copies_available	author_id
▶	109	The God of Small Things	Fiction	1997	3	9
	116	Malgudi Days	Fiction	1943	4	16

Insight: useful for finding old/classic fiction books.

IN LIST

Question: - Retrieve books that belong to the genre "mystery", "horror", "historical".

Input Query:

```
select * from books
where genre in("mystery","horror","historical");
```

Output:

	book_id	title	genre	published_year	copies_available	author_id
▶	104	Murder on the Orient Express	Mystery	1934	2	4
	111	It	Horror	1986	2	11
	114	War and Peace	Historical	1869	2	14
	120	Sherlock Holmes: A Study in Scarlet	Mystery	1887	4	20

Insight: used when filtering multiple values.

LIKE

Question: -Find all books where the title starts with "the".

Input Query:

```
select * from books
where title like "The%";
```

Output:

	book_id	title	genre	published_year	copies_available	author_id
▶	105	The Alchemist	Philosophy	1988	6	5
	106	The Da Vinci Code	Thriller	2003	7	6
	108	The Hobbit	Fantasy	1937	5	8
	109	The God of Small Things	Fiction	1997	3	9
	110	The Kite Runner	Drama	2003	4	10
	115	The Picture of Dorian Gray	Philosophy	1890	3	15
	117	The Old Man and the Sea	Drama	1952	3	17

Insight: demonstrates partial text matching.

GROUP BY WITH HAVING:

Question: Retrieving the total books based on genre where the count is greater than 2.

Input Query:

```
select genre, count(*) as total_books from books
group by genre
having total_books>2;
```

Output:

	genre	total_books
▶	Fantasy	3
	Fiction	3
	Classic	3

Insights: This helps to identify popular categories of books.

ORDER BY & LIMIT:

Question: finding the latest 5 member that has joined

Input Query:

```
select * from members
order by join_date desc
limit 5;
```

Output:

	member_id	first_name	last_name	email	join_date	phone
▶	208	Neha	Iyer	neha.iyer@example.com	2025-08-30	9456789031
	202	Priya	Verma	priya.verma@example.com	2025-06-20	8765432109
	215	Aditya	Roy	aditya.roy@example.com	2024-02-14	9123456708
	214	Meera	Joshi	meera.joshi@example.com	2023-10-01	9012345697
	213	Arjun	Khanna	arjun.khanna@example.com	2023-07-18	9901234586

Insight: sorting by members joining date to identify the latest 5 members that have joined.

JOINS:

Question: finding the books published between the year 2000 to 2010

Input Query:

```
select k.title, a.name, k.published_year from books k
join authors a
on k.author_id=a.author_id
where k.published_year between 2000 and 2010;
```

Output

	title	name	published_year
▶	Five Point Someone	Chetan Bhagat	2004
	The Da Vinci Code	Dan Brown	2003
	The Kite Runner	Khaled Hosseini	2003

Insights: Helps to analyze books from a specific decade.

Question: retrieving title of books and their authors that belongs to the country "UK".

Input Query:

```
select b.Title, a.Name from authors a
join books b
on a.author_id=b.author_id
where a.country='uk';
```

Output:

	Title	Name
▶	Harry Potter and the Sorcerer's Stone	J.K. Rowling
	Murder on the Orient Express	Agatha Christie
	The Hobbit	J.R.R. Tolkien
	Pride and Prejudice	Jane Austen
	Oliver Twist	Charles Dickens
	Sherlock Holmes: A Study in Scarlet	Arthur Conan Doyle

Insights: It helps to filter books based on the author's nationality.

NESTED JOINS:

Question: Retrieving the information from 3 tables

Input Query:

```
select concat(m.first_name," ", m.last_name) as full_name, k.book_id,
k.title, b.borrow_date, b.return_date from borrowings b
join members m
on b.member_id=m.member_id
join books k
on b.book_id= k.book_id;
```

Output:

	full_name	book_id	title	borrow_date	return_date
▶	Priya Verma	107	To Kill a Mockingbird	2020-07-01	2020-07-15
	Aditya Roy	115	The Picture of Dorian Gray	2024-02-20	NULL
	Sneha Kapoor	101	Harry Potter and the Sorcerer's Stone	2021-06-01	NULL
	Rahul Desai	110	The Kite Runner	2022-04-10	2022-04-25
	Siddharth Rao	104	Murder on the Orient Express	2023-03-15	2023-03-30
	Amit Sharma	117	The Old Man and the Sea	2020-03-20	2020-04-05
	Arjun Khanna	109	The God of Small Things	2023-07-20	2023-08-05
	Divya Malhotra	103	Five Point Someone	2023-01-25	2023-02-10
	Vikram Singh	112	Adventures of Tom Sawyer	2021-09-20	NULL
	Meera Joshi	108	The Hobbit	2023-10-05	2023-10-20
	Anjali Nair	102	A Game of Thrones	2022-02-25	2022-03-12
	Karan Patel	111	It	2022-11-20	2022-12-05
	Pooja Sharma	105	The Alchemist	2023-06-10	2023-06-25
	Rohan Mehta	114	War and Peace	2021-01-15	NULL
	Neha Tver	119	One Hundred Years of Solitude	2022-09-01	2022-09-15

Insights: Combines multiple tables to show title & which member borrowed which book and their return date.

IS NULL / IS NOT NULL

Question: list the member's name and title where books are currently borrowed and not returned.

Input Query:

```
select concat(m.first_name," ",m.last_name) as full_name,
k.title, m.phone,m.email from borrowings b
join members m
on b.member_id=m.member_id
join books k
on b.book_id=k.book_id
where b.return_date is null;
```

Output:

	full_name	title	phone	email
►	Aditya Roy	The Picture of Dorian Gray	9123456708	aditya.roy@example.com
	Sneha Kapoor	Harry Potter and the Sorcerer's Stone	6543210987	sneha.kapoor@example.com
	Vikram Singh	Adventures of Tom Sawyer	9123456780	vikram.singh@example.com
	Rohan Mehta	War and Peace	7654321098	rohan.mehta@example.com

Insight: This query help to analyse which member has not return the book.

Question: calculating which 5 members has kept book for the longest time

Input Query:

```
select b.borrow_id, concat(m.first_name, m.last_name) as full_name, k.title,
datediff(b.return_date,b.borrow_date) as days_taken from borrowings b
join members m
on b.member_id=m.member_id
join books k
on b.book_id=k.book_id
where b.return_date is not null
order by days_taken desc
limit 5;
```

Output:

	borrow_id	full_name	title	days_taken
▶	306	AmitSharma	The Old Man and the Sea	16
	307	ArjunKhanna	The God of Small Things	16
	308	DivyaMalhotra	Five Point Someone	16
	304	RahulDesai	The Kite Runner	15
	305	SiddharthRao	Murder on the Orient Express	15

Insights: Useful for monitoring late returns.

SUBQUERIES:

Question: Retrieving the books which has more books copy available than the average.

Input Query:

```
select title, copies_available from books
where copies_available > (select avg(copies_available) from books);
```

Output:

	title	copies_available
▶	The Da Vinci Code	7
	The Alchemist	6
	Harry Potter and the Sorcerer's Stone	5
	The Hobbit	5
	Five Point Someone	4
	To Kill a Mockingbird	4
	The Kite Runner	4
	Pride and Prejudice	4
	Malgudi Days	4
	Sherlock Holmes: A Study in Scarlet	4

Insight: Retrieves books that have more copies than the average availability.

Question: Find country with the most authors

Input Query:

```
select name, country from authors
where country = (select country from authors
group by country
order by count(*) desc
limit 1);
```

Output:

	name	country
▶	J.K. Rowling	UK
	Agatha Christie	UK
	J.R.R. Tolkien	UK
	Jane Austen	UK
	Charles Dickens	UK
	Arthur Conan Doyle	UK

Insight: This query helps to retrieve the country with most authors.

Question: Find the country of the author who wrote the oldest published book.

Input Query:

```
select a.name, a.country, b.published_year from authors a
join books b
on a.author_id=b.author_id
where b.published_year=(select min(published_year) from books);
```

Output:

	name	country	published_year
▶	Jane Austen	UK	1813

Insights: by combining 2 tables fetching the oldest published book in the database.

WINDOWS FUNCTION:

ROW NUMBER:

Question: Assign row numbers to books by published year.

Input Query:

```
select book_id, title, published_year,
row_number() over( order by published_year asc) as row_num from books;
```

Output:

	book_id	title	published_year	row_num
▶	113	Pride and Prejudice	1813	1
	118	Oliver Twist	1838	2
	114	War and Peace	1869	3
	112	Adventures of Tom Sawyer	1876	4
	120	Sherlock Holmes: A Study in Scarlet	1887	5
	115	The Picture of Dorian Gray	1890	6
	104	Murder on the Orient Express	1934	7
	108	The Hobbit	1937	8
	116	Malgudi Days	1943	9
	117	The Old Man and the Sea	1952	10
	107	To Kill a Mockingbird	1960	11
	119	One Hundred Years of Solitude	1967	12
	111	It	1986	13
	105	The Alchemist	1988	14
	102	A Game of Thrones	1996	15

Insight: Gives a unique row number to books ordered by published year.

DENSE_RANK:

Question: Rank books by how recently they were published with DENSE_RANK.

Input Query:

```
select title, published_year,
dense_rank() over(order by published_year desc) as recent_rank
from books;
```

Output

	title	published_year	recent_rank
▶	Five Point Someone	2004	1
	The Da Vinci Code	2003	2
	The Kite Runner	2003	2
	Harry Potter and the Sorcerer's Stone	1997	3
	The God of Small Things	1997	3
	A Game of Thrones	1996	4
	The Alchemist	1988	5
	It	1986	6
	One Hundred Years of Solitude	1967	7
	To Kill a Mockingbird	1960	8
	The Old Man and the Sea	1952	9
	Malgudi Days	1943	10
	The Hobbit	1937	11
	Murder on the Orient Express	1934	12
	The Picture of Dorian Gray	1890	13

Insight: Ranks books by how recent they are, without skipping numbers for ties.

Question: Rank books by how recently they were published with RANK.

INPUT QUERY:

```
select title, published_year,
rank() over(order by published_year desc) as recent_rank
from books;
```

OUTPUT:

	title	published_year	recent_rank
►	Five Point Someone	2004	1
	The Da Vinci Code	2003	2
	The Kite Runner	2003	2
	Harry Potter and the Sorcerer's Stone	1997	4
	The God of Small Things	1997	4
	A Game of Thrones	1996	6
	The Alchemist	1988	7
	It	1986	8
	One Hundred Years of Solitude	1967	9
	To Kill a Mockingbird	1960	10
	The Old Man and the Sea	1952	11
	Malgudi Days	1943	12
	The Hobbit	1937	13
	Murder on the Orient Express	1934	14
	The Picture of Dorian Gray	1890	15

Insight: Similar to DENSE_RANK but skips ranks if multiple books share the same year.